rtl/

Dragon Core RTL Architecture - Full Configuration

Complete architecture: RTL-001 through RTL-024; five functional domains plus top-level integration

top / integration

RTL-024 - Dragon Core Top-Level Integration

SPI bridge, adapters, clocks, resets, pads, module assembly and technology boundary.

instantiates and integrates RTL-001 through RTL-023

infrastructure /

RTL-001 - Configuration Register Bank

Central register endpoint, decode, protection, configuration, commands, status and proxy access.

RTL-002 - System Control Logic

Startup sequence, operating state, enables, resets, readiness and degraded/fault coordination.

RTL-003 - Interrupt Controller

32-source interrupt latching, masking, W1C, priority and aggregate output.

RTL-004 - Timestamp Engine

64-bit monotonic timebase, snapshot, distribution and synchronization support.

RTL-005 - Event Logger

Event arbitration, 64-record storage, overflow and host retrieval.

RTL-006 - GPIO Controller

Eight GPIO channels, qualification, data, direction, events and interrupts.

monitoring /

RTL-007 - Rotor RPM Monitor

Rotor edge capture, measurement, thresholds, status, events and interrupts.

RTL-008 - Engine RPM Monitor

Engine edge capture, measurement, thresholds, status, events and interrupts.

RTL-009 - Pulse/Frequency Measurement

Two general pulse channels, qualification and frequency measurement.

RTL-010 - Sensor Monitor Interface

Sampled values, validity, age, stale detection and thresholds.

RTL-011 - System Health Monitor

Eight conditioned inputs, qualification, status, faults, events and interrupts.

safety /

RTL-012 - Watchdog Manager

External service supervision, timeout, evidence and authorized response request.

RTL-013 - Fault Manager

Fault vector, class mapping, latching, degraded evidence and response coordination.

RTL-014 - Interlock Engine

Finite interlock expressions, guarded actions and inhibit status.

RTL-015 - Lost-Link Supervisor

Activity supervision, timeout, recovery and authorized response request.

RTL-016 - Safety Output Controller

Sole logical safety-output owner and final internal safety gating.

databus /

RTL-017 - ARINC 429 Receive Interface

Predecoded word buffering, timestamping, status, events, interrupts and routing source.

RTL-018 - ARINC 429 Transmit Interface

TX buffering, parity, serialization, status, events, interrupts and routing destination.

RTL-019 - CAN Interface

Classical CAN protocol, buffers, status, events, interrupts and routing.

RTL-020 - UART Interface

UART RX/TX protocol, buffers, status, events, interrupts and routing.

RTL-021 - Databus Routing Engine

Finite message route authorization, arbitration, replication and destination queues.

user logic /

RTL-022 - Mask-Defined User Logic Island

Fixed four-equation user logic, two four-word mailboxes, four user I/O pads, deterministic transforms and guarded notifications.

RTL-023 - User-Configurable Routing Logic

Finite approved signal-endpoint routing with protection, validation and diagnostics.

Module names and responsibilities derived from RCA-DFC-ARCH-001 Rev F Released_Final.